

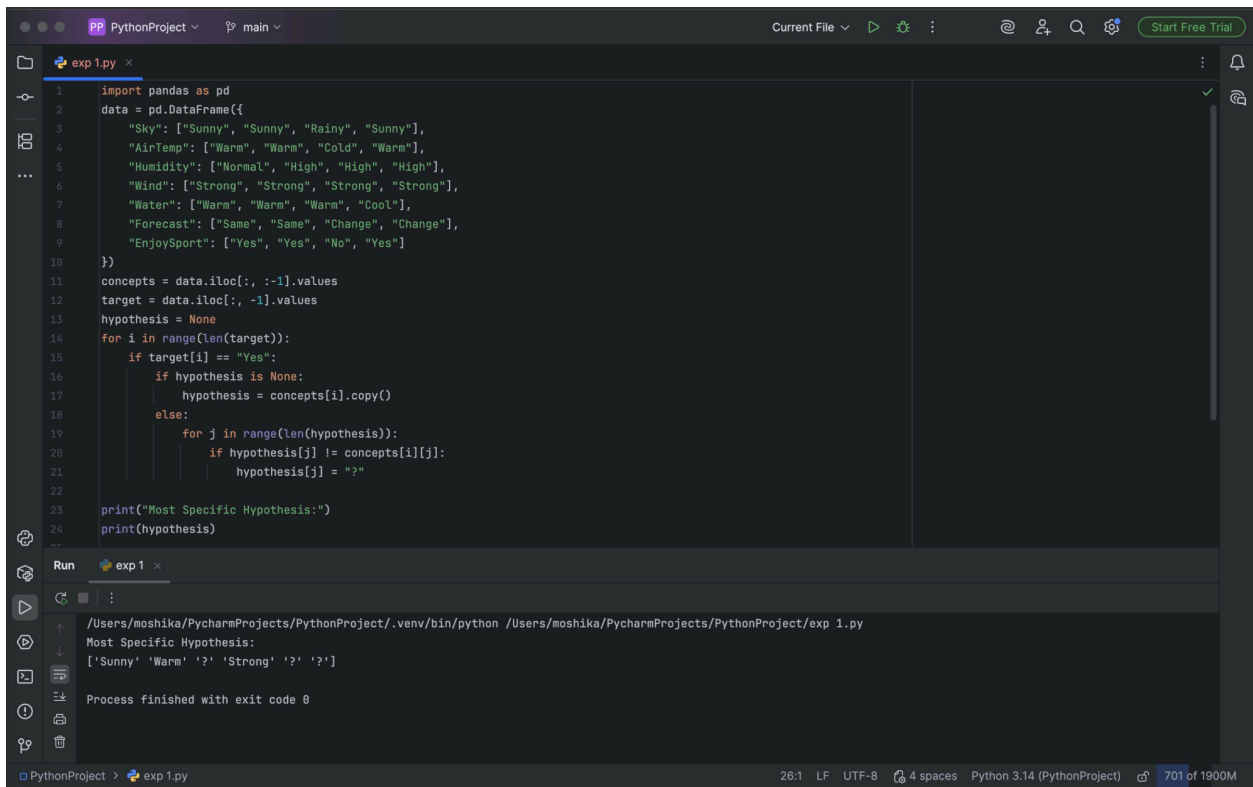
EXPERIMENT - 1

Aim: To find the most specific hypothesis that is consistent with the positive training examples.

Procedure:

1. Load the training dataset.
2. Initialize the hypothesis with the most specific values.
3. Read each training example one by one.
4. Check whether the example is positive.
5. If positive, compare it with the current hypothesis.
6. Generalize the hypothesis where attributes differ.
7. Ignore negative examples.
8. Display the final most specific hypothesis.

Output:



```
1 import pandas as pd
2 data = pd.DataFrame({
3     "Sky": ["Sunny", "Sunny", "Rainy", "Sunny"],
4     "AirTemp": ["Warm", "Warm", "Cold", "Warm"],
5     "Humidity": ["Normal", "High", "High", "High"],
6     "Wind": ["Strong", "Strong", "Strong", "Strong"],
7     "Water": ["Warm", "Warm", "Warm", "Cool"],
8     "Forecast": ["Same", "Same", "Change", "Change"],
9     "EnjoySport": ["Yes", "Yes", "No", "Yes"]
10 })
11 concepts = data.iloc[:, :-1].values
12 target = data.iloc[:, -1].values
13 hypothesis = None
14 for i in range(len(target)):
15     if target[i] == "Yes":
16         if hypothesis is None:
17             hypothesis = concepts[i].copy()
18         else:
19             for j in range(len(hypothesis)):
20                 if hypothesis[j] != concepts[i][j]:
21                     hypothesis[j] = "?"
22
23 print("Most Specific Hypothesis:")
24 print(hypothesis)
```

Run exp 1 x

```
/Users/moshika/PycharmProjects/PythonProject/.venv/bin/python /Users/moshika/PycharmProjects/PythonProject/exp 1.py
Most Specific Hypothesis:
['Sunny' 'Warm' '?' 'Strong' '?' '?']

Process finished with exit code 0
```

PythonProject > exp 1.py 26:1 LF UTF-8 4 spaces Python 3.14 (PythonProject) 701 of 1900M

Result: The FIND-S algorithm successfully finds the most specific hypothesis consistent with the given training data.